**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 095 · Lesson 095 · CPU Pipelining: Overlap and Hazards

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is pipeline throughput?

Q2. What is a hazard?

Q3. Which CPU feature may reduce RAM accesses?

Q4. A D flip-flop has D=0 and Q=1 just before the active rising edge. What is Q after?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

pipeline — Stages overlapping different instructions.

throughput — Completed work per time.

data hazard — Dependency on unavailable result.

control hazard — Branch uncertainty interrupting fetch.

## New learning

Observe — Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

Explain — With four independent instructions and three one-cycle stages, ideal completion is $3+4-1=6$ cycles; serial non-overlap takes $4\times3=12$.

Evaluate — Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

## Worked example

Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.

1. Explain the mechanism — $2+3-1=4$ cycles; serial baseline $2\times3=6$ cycles. — The second task starts stage one while first advances to stage two.

# We Do · Practise with guidance

Five independent instructions use a four-stage ideal pipeline. Calculate completion.

Explain the mechanism

$4+5-1=8$ cycles, compared with $4\times5=20$ serial stage cycles.

Assumes equal one-cycle stages, no resource conflicts or stalls.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Show cycles for I1-I3 in a 3-stage F-D-E pipeline, and finish cycle of I3. [6]

### Question 2

Compute ideal pipeline versus serial completion for 6 instructions and 3 stages. [6]

### Question 3

Explain why branch and data hazards can reduce actual improvement, distinguishing latency and throughput. [6]

# Exit Ticket · Answer before checking

A pipeline makes each stage finish in less than one cycle by default.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Rate of completed instructions.

Q2. Dependency/resource/control issue that interrupts pipeline.

Q3. Cache.

Q4. Q becomes0, sampled at the active edge.

### You Do pages · Independent questions

Q1. I1 F1 D2 E3 [2], I2 F2 D3 E4 [2], I3 F3 D4 E5 [2].

Q2. Pipeline3+6−1=8 cycles [3], serial6×3=18 [2], equal stage/no stall assumption [1].

Q3. Data dependency waiting [2], branches flush/stall [2], average throughput drops though per-instruction latency not necessarily improved [2].

### Exit Ticket · Model answer

False. It overlaps stages, improving throughput when conditions allow.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**